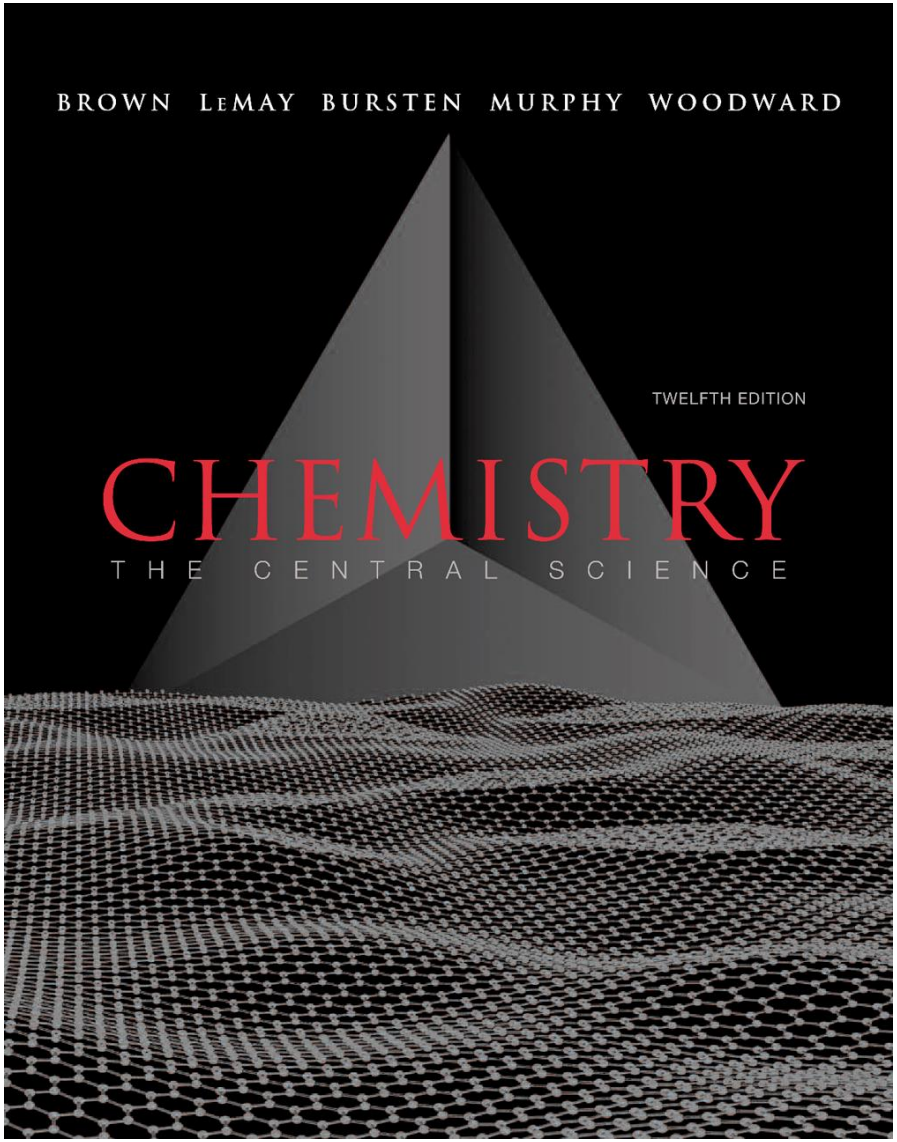


Lecture Presentation

Chapter 10

Gases

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Characteristics of Gases

- Unlike liquids and solids, gases
  - Expand to fill their containers.
  - Are highly compressible.
  - Have extremely low densities.

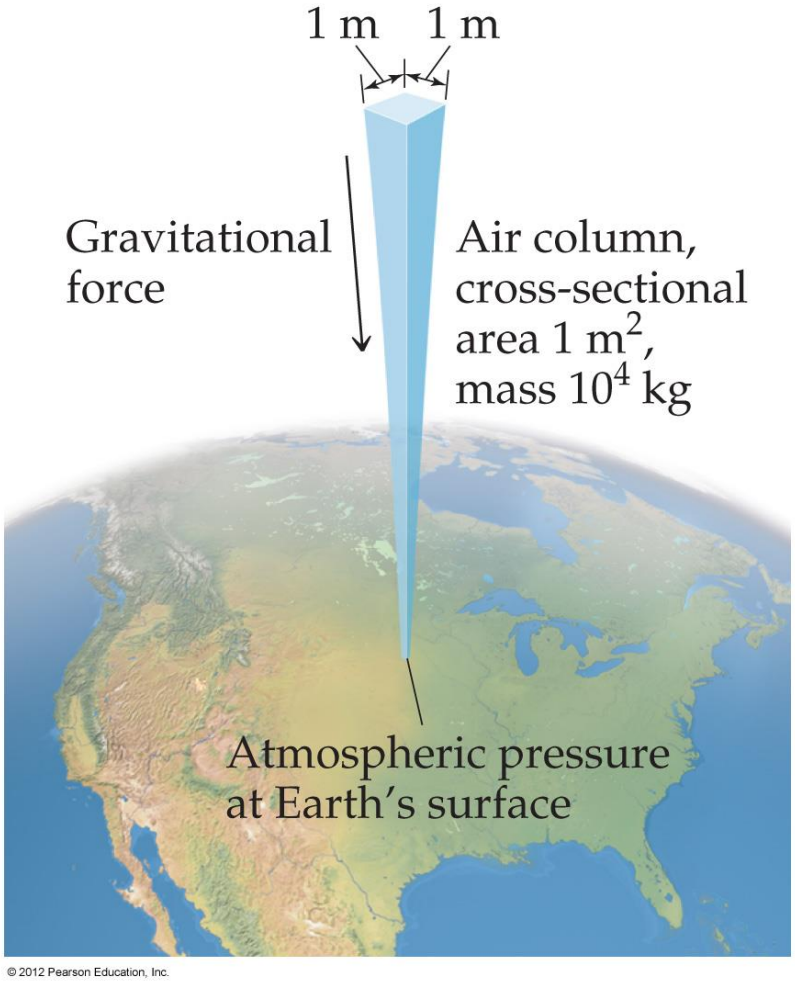
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Pressure

- **Pressure** is the amount of force applied to an area:

$$P = \frac{F}{A}$$

- **Atmospheric pressure** is the weight of air per unit of area.



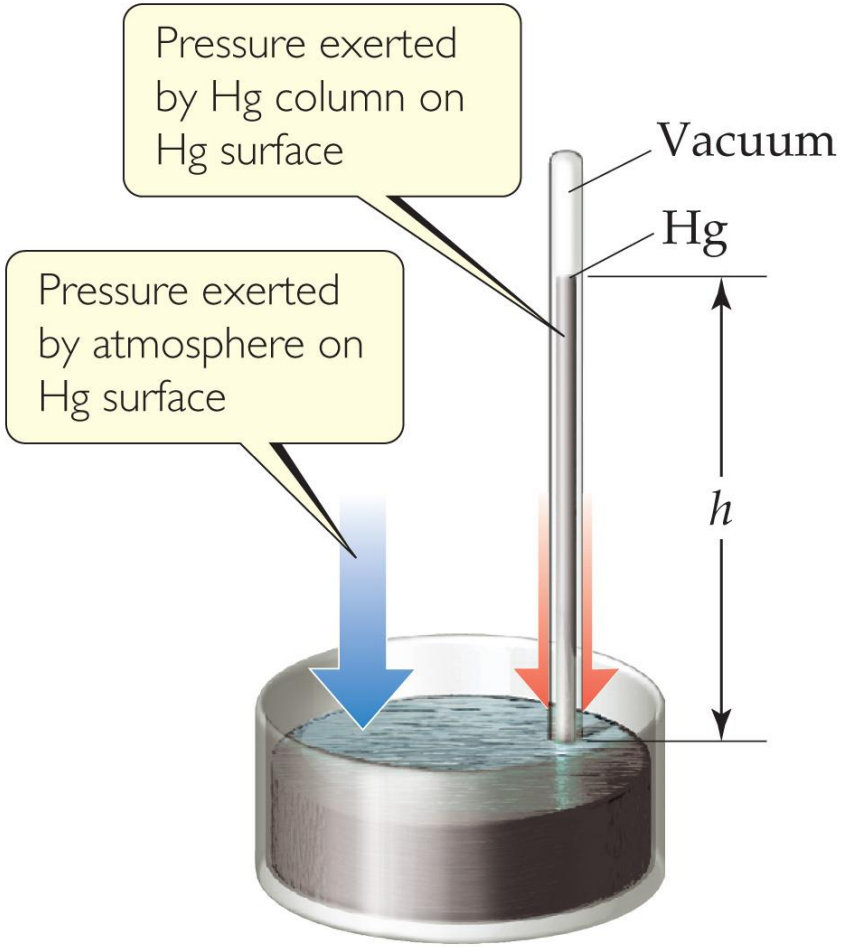
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Units of Pressure

- Pascals
  - 1 Pa = 1 N/m^2
- Bar
  - 1 bar = 10^5 Pa = 100 kPa

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Units of Pressure

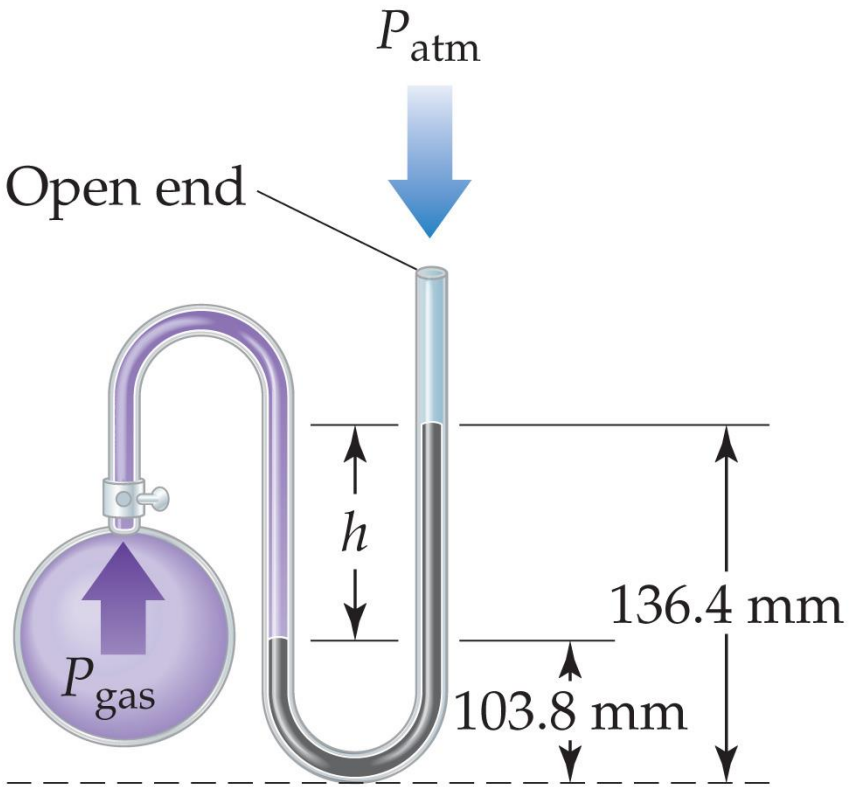


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- mmHg or torr
  - These units are literally the difference in the heights measured in mm (h) of two connected columns of mercury.
- Atmosphere
  - 1.00 atm = 760 torr

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Manometer



$$P_{gas} = P_{atm} + P_h$$

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The manometer is used to measure the difference in pressure between atmospheric pressure and that of a gas in a vessel.

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Standard Pressure

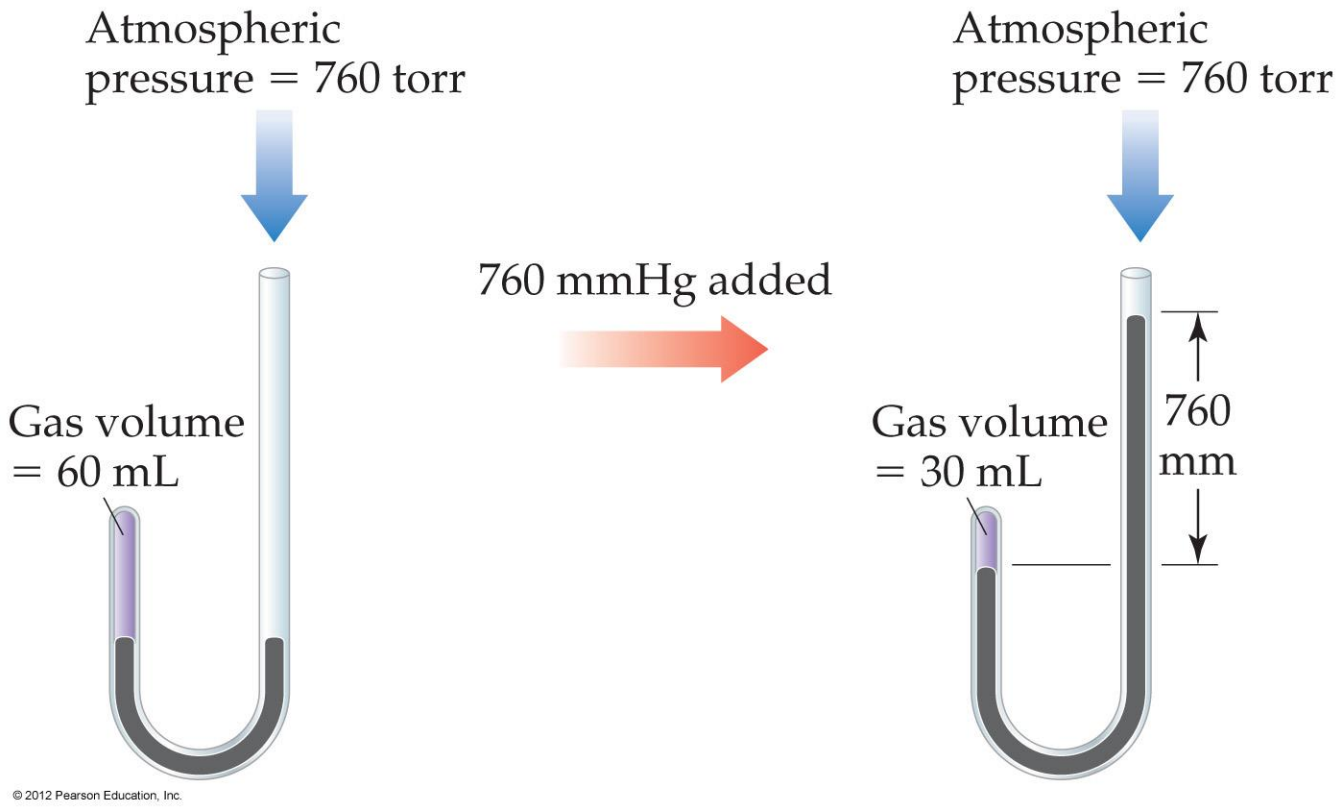
- Normal atmospheric pressure at sea level is referred to as **standard pressure**.

- It is equal to
  - 1.00 atm
  - 760 torr (760 mmHg)
  - 101.325 kPa

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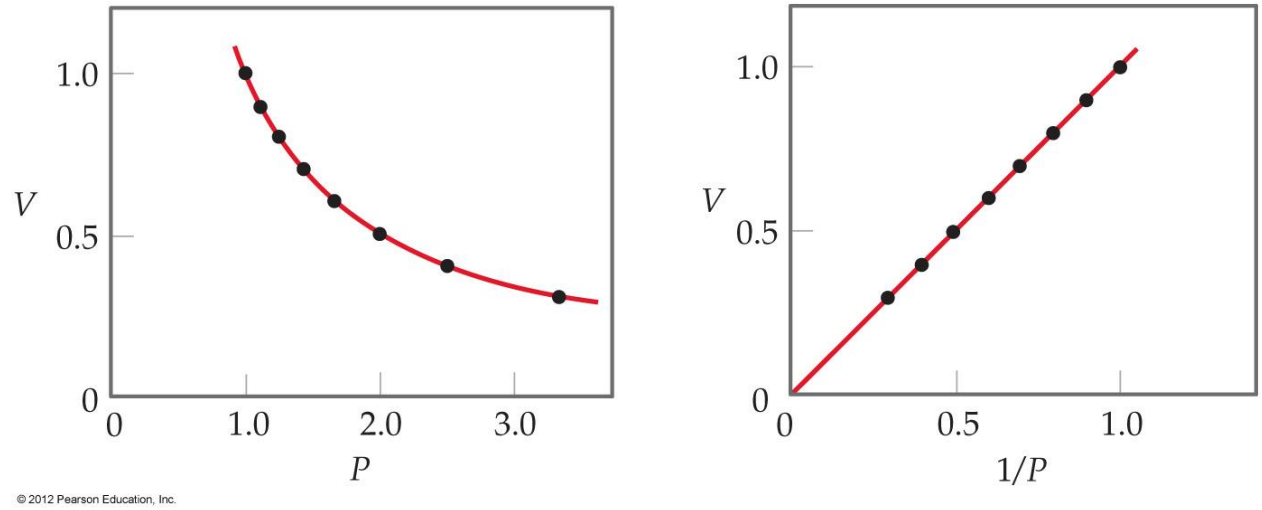
Boyle's Law

The volume of a fixed quantity of gas at constant temperature is inversely proportional to the pressure.



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P and *V* are Inversely Proportional



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Since $PV = k$

A plot of *V* versus *P* results in a curve.

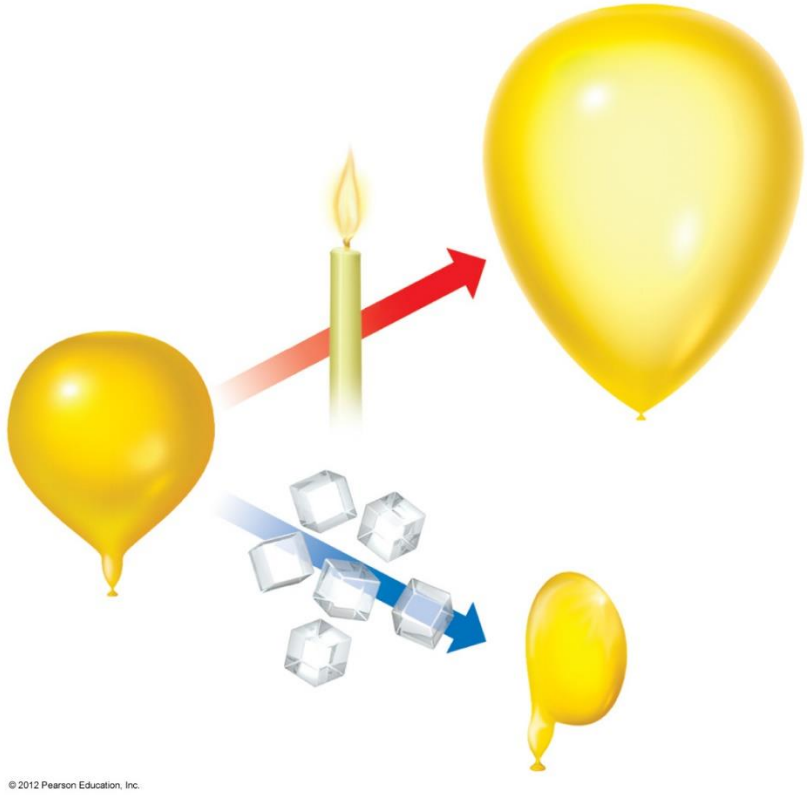
$$V = k\,(1/P)$$

This means a plot of *V* versus 1/*P* will be a straight line.

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Charles's Law

- The volume of a fixed amount of gas at constant pressure is directly proportional to its absolute temperature.

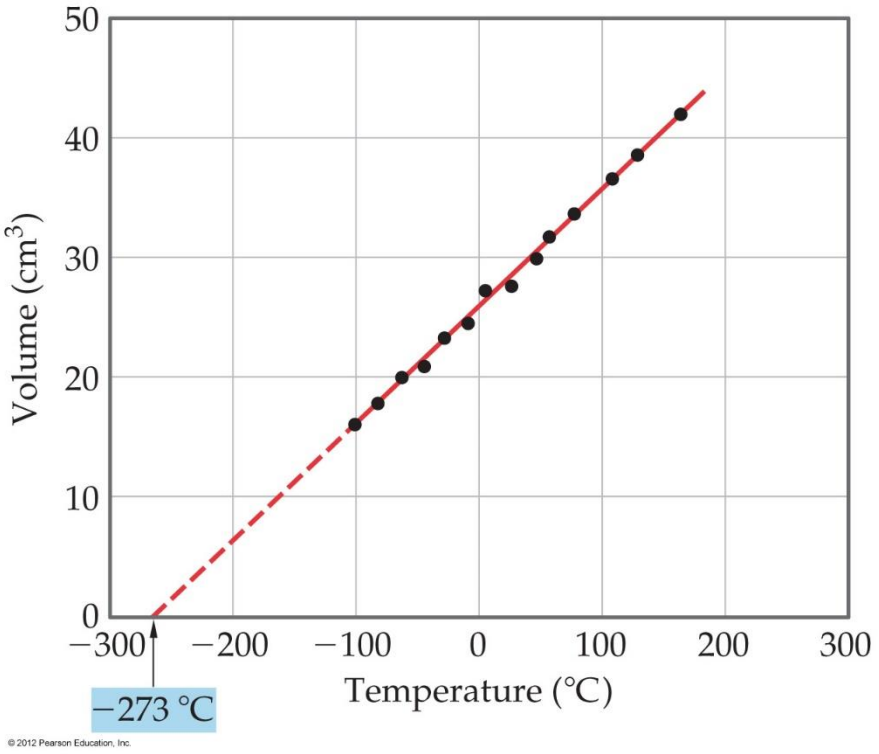


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Charles's Law

- So, $\frac{V}{T} = k$

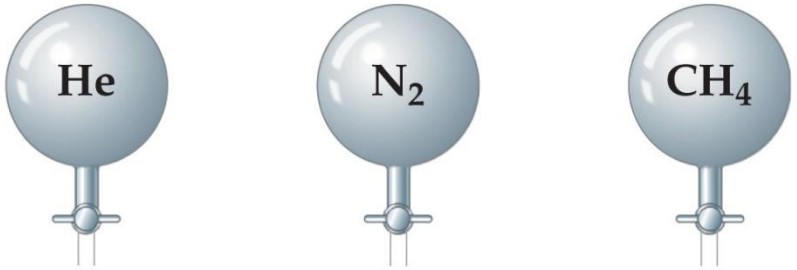
- A plot of *V* versus *T* will be a straight line.



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Avogadro's Law

- The volume of a gas at constant temperature and pressure is directly proportional to the number of moles of the gas.
- Mathematically, this means $V = kn$



Volume	22.4 L	22.4 L	22.4 L
Pressure	1 atm	1 atm	1 atm
Temperature	0 °C	0 °C	0 °C
Mass of gas	4.00 g	28.0 g	16.0 g
Number of gas molecules	6.02×10^{23}	6.02×10^{23}	6.02×10^{23}

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Gases

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The conditions 0 °C and 1 atm are referred to as **standard temperature and pressure (STP)**.
The volume occupied by 1 mol of ideal gas at STP, 22.41 L, is known as the volume of an ideal gas at STP.

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Ideal-Gas Equation

- So far we've seen that

$V \propto 1/P$ (Boyle's law)

$V \propto T$ (Charles's law)

$V \propto n$ (Avogadro's law)

- Combining these, we get

$$V \propto \frac{nT}{P}$$

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Ideal-Gas Equation

The constant of proportionality is known as *R*, the gas constant.

TABLE 10.2 • Numerical Values of the Gas Constant *R* in Various Units

Units	Numerical Value
L-atm/mol-K	0.08206
J/mol-K*	8.314
cal/mol-K	1.987
m^3-Pa/mol-K*	8.314
L-torr/mol-K	62.36

*SI unit

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Ideal-Gas Equation

The relationship

$$V \propto \frac{nT}{P}$$

then becomes

$$V = R\,\frac{nT}{P}$$

or

$$PV = nRT$$

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Densities of Gases

If we divide both sides of the ideal-gas equation by V and by RT, we get

$$\frac{n}{V} = \frac{P}{RT}$$

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Densities of Gases

- We know that
  - Moles $\times$ molecular mass = mass

$$n \times M = m$$

- So multiplying both sides by the molecular mass (M) gives

$$\frac{m}{V} = \frac{PM}{RT}$$

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Densities of Gases

- Mass ÷ volume = density
- So, $$d = \frac{m}{V} = \frac{PM}{RT}$$

Note: One needs to know only the molecular mass, the pressure, and the temperature to calculate the density of a gas.

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Molecular Mass

We can manipulate the density equation to enable us to find the molecular mass of a gas:

$$d = \frac{PM}{RT}$$

becomes

$$M = \frac{dRT}{P}$$

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Dalton's Law of Partial Pressures

- The total pressure of a mixture of gases equals the sum of the pressures that each would exert if it were present alone.
- In other words,

$$P_{\text{total}} = P_1 + P_2 + P_3 + \ldots$$

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If we let P_t be the total pressure of a mixture of gases and P_1, P_2, P_3, and so forth be the partial pressures of the individual gases, we can write Dalton's law of partial pressures as

$$P_t = P_1 + P_2 + P_3 + \ldots \qquad [10.12]$$

This equation implies that each gas behaves independently of the others, as we can see by the following analysis. Let n_1, n_2, n_3, and so forth be the number of moles of each of the gases in the mixture and n_t be the total number of moles of gas. If each gas obeys the ideal-gas equation, we can write

$$P_1 = n_1\left(\frac{RT}{V}\right); \quad P_2 = n_2\left(\frac{RT}{V}\right); \quad P_3 = n_3\left(\frac{RT}{V}\right); \quad \text{and so forth}$$

All of the gases in a container must occupy the same volume and will come to the same temperature in a relatively short period of time. Using these facts to simplify Equation 10.12, we obtain

$$P_t = (n_1 + n_2 + n_3 + \ldots)\left(\frac{RT}{V}\right) = n_t\left(\frac{RT}{V}\right) \qquad [10.13]$$

That is, at constant temperature and constant volume the total pressure of a gas sample is determined by the total number of moles of gas present, whether that total represents just one gas or a mixture of gases.

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SAMPLE EXERCISE 10.10 Applying Dalton's Law of Partial Pressures

A mixture of 6.00 g of $O_2(g)$ and 9.00 g of $CH_4(g)$ is placed in a 15.0-L vessel at 0 °C. What is the partial pressure of each gas, and what is the total pressure in the vessel?

SOLUTION

Analyze We need to calculate the pressure for two gases in the same volume and at the same temperature.

Plan Because each gas behaves independently, we can use the ideal-gas equation to calculate the pressure each would exert if the other were not present. Per Dalton's law, the total pressure is the sum of these two partial pressures.

Solve We first convert the mass of each gas to moles:

$$n_{O_2} = (6.00\text{ g } O_2)\left(\frac{1\text{ mol } O_2}{32.0\text{ g } O_2}\right) = 0.188\text{ mol } O_2$$

$$n_{CH_4} = (9.00\text{ g } CH_4)\left(\frac{1\text{ mol } CH_4}{16.0\text{ g } CH_4}\right) = 0.563\text{ mol } CH_4$$

We use the ideal-gas equation to calculate the partial pressure of each gas:

$$P_{O_2} = \frac{n_{O_2}RT}{V} = \frac{(0.188\text{ mol})(0.08206\text{ L-atm/mol-K})\ (273\text{ K})}{15.0\text{ L}} = 0.281\text{ atm}$$

$$P_{CH_4} = \frac{n_{CH_4}RT}{V} = \frac{(0.563\text{ mol})(0.08206\text{ L-atm/mol-K})(273\text{ K})}{15.0\text{ L}} = 0.841\text{ atm}$$

According to Dalton's law of partial pressures (Equation 10.12), the total pressure in the vessel is the sum of the partial pressures:

$$P_t = P_{O_2} + P_{CH_4} = 0.281\text{ atm} + 0.841\text{ atm} = 1.122\text{ atm}$$

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Partial Pressures and Mole Fractions

Because each gas in a mixture behaves independently, we can relate the amount of a given gas in a mixture to its partial pressure. For an ideal gas, we can write

$$\frac{P_1}{P_t} = \frac{n_1 RT/V}{n_t RT/V} = \frac{n_1}{n_t} \qquad [10.14]$$

The ratio n_1/n_t is called the *mole fraction of gas 1*, which we denote X_1. The **mole fraction**, X, is a dimensionless number that expresses the ratio of the number of moles of one component in a mixture to the total number of moles in the mixture. Thus, for gas 1 we have

$$X_1 = \frac{\text{Moles of compound 1}}{\text{Total moles}} = \frac{n_1}{n_t} \qquad [10.15]$$

We can combine Equations 10.14 and 10.15 to give

$$P_1 = \left(\frac{n_1}{n_t}\right) P_t = X_1 P_t \qquad [10.16]$$

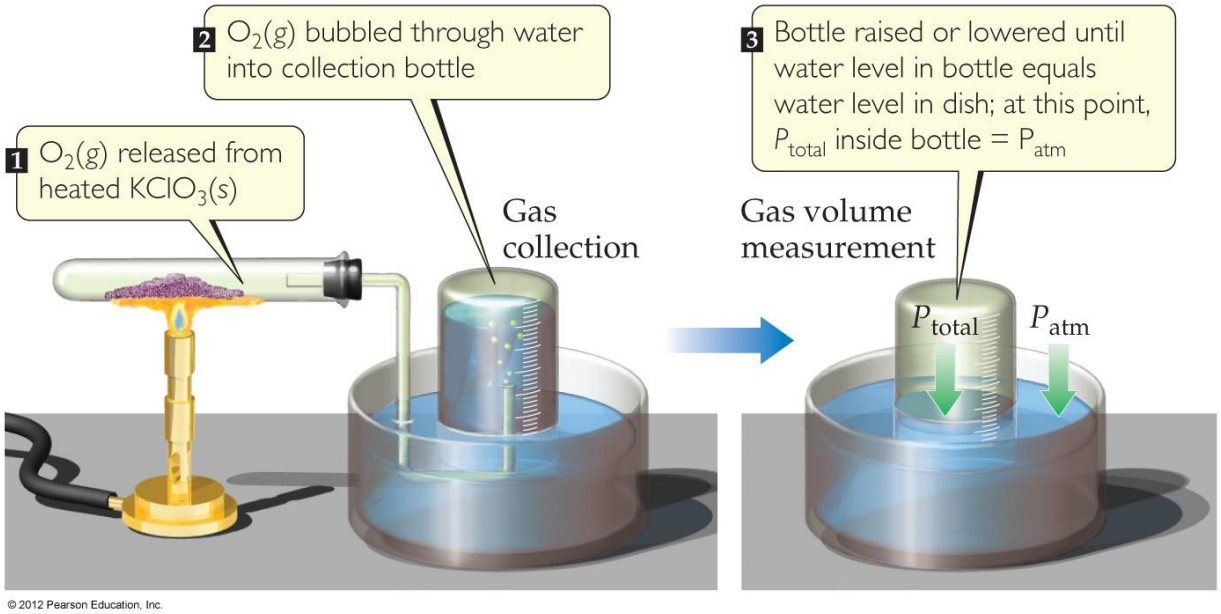
The mole fraction of N_2 in air is 0.78—that is, 78% of the molecules in air are N_2. This means that if the barometric pressure is 760 torr, the partial pressure of N_2 is

$$P_{N_2} = (0.78)(760 \text{ torr}) = 590 \text{ torr}$$

This result makes intuitive sense: Because N_2 makes up 78% of the mixture, it contributes 78% of the total pressure.

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Partial Pressures

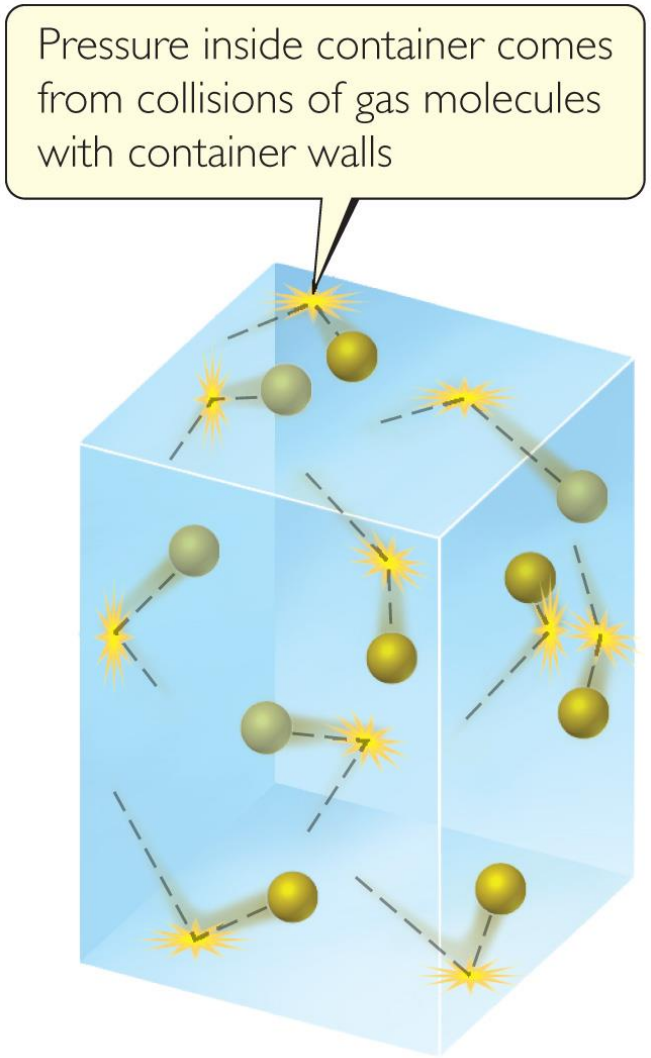


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- When one collects a gas over water, there is water vapor mixed in with the gas.
- To find only the pressure of the desired gas, one must subtract the vapor pressure of water from the total pressure.

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Kinetic-Molecular Theory



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This is a model that aids in our understanding of what happens to gas particles as environmental conditions change.

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Main Tenets of Kinetic-Molecular Theory

Gases consist of large numbers of molecules that are in continuous, random motion.

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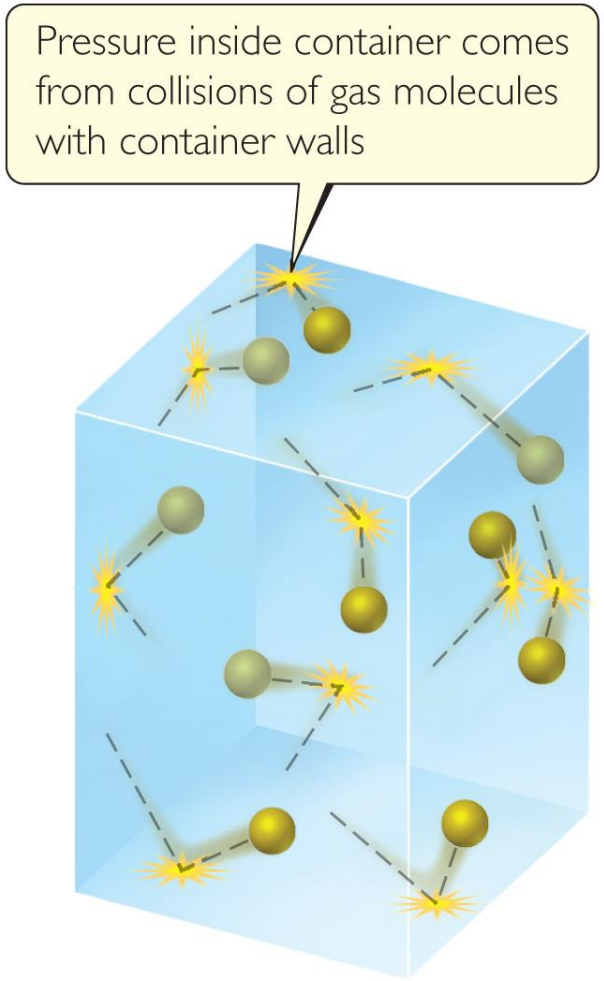
Main Tenets of Kinetic-Molecular Theory

The combined volume of all the molecules of the gas is negligible relative to the total volume in which the gas is contained.

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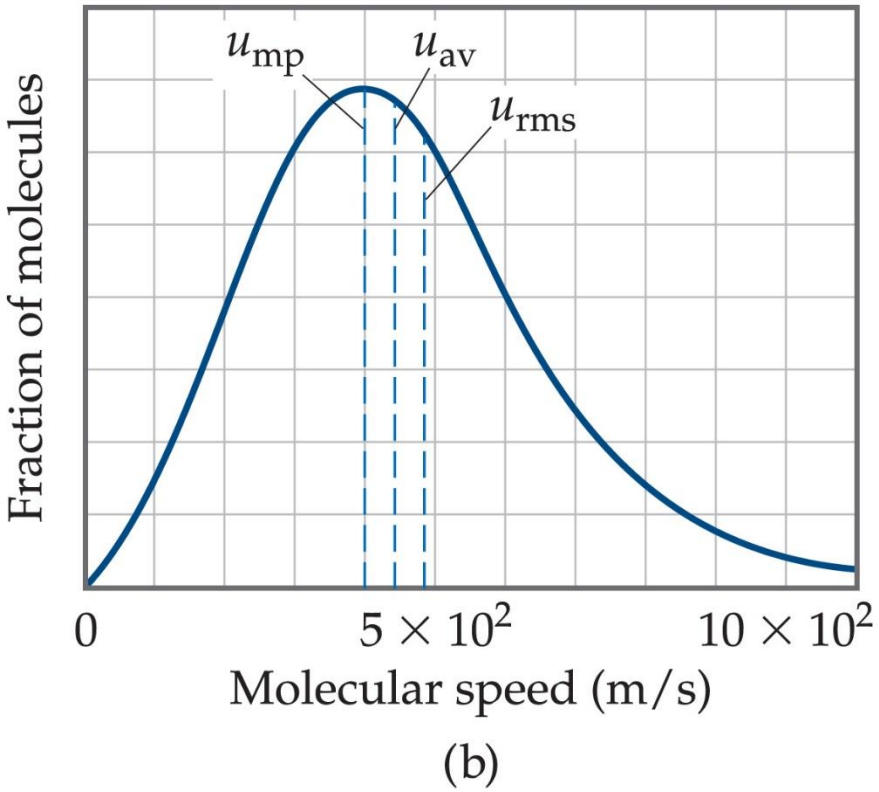
Main Tenets of Kinetic-Molecular Theory

Attractive and repulsive forces between gas molecules are negligible.



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Main Tenets of Kinetic-Molecular Theory



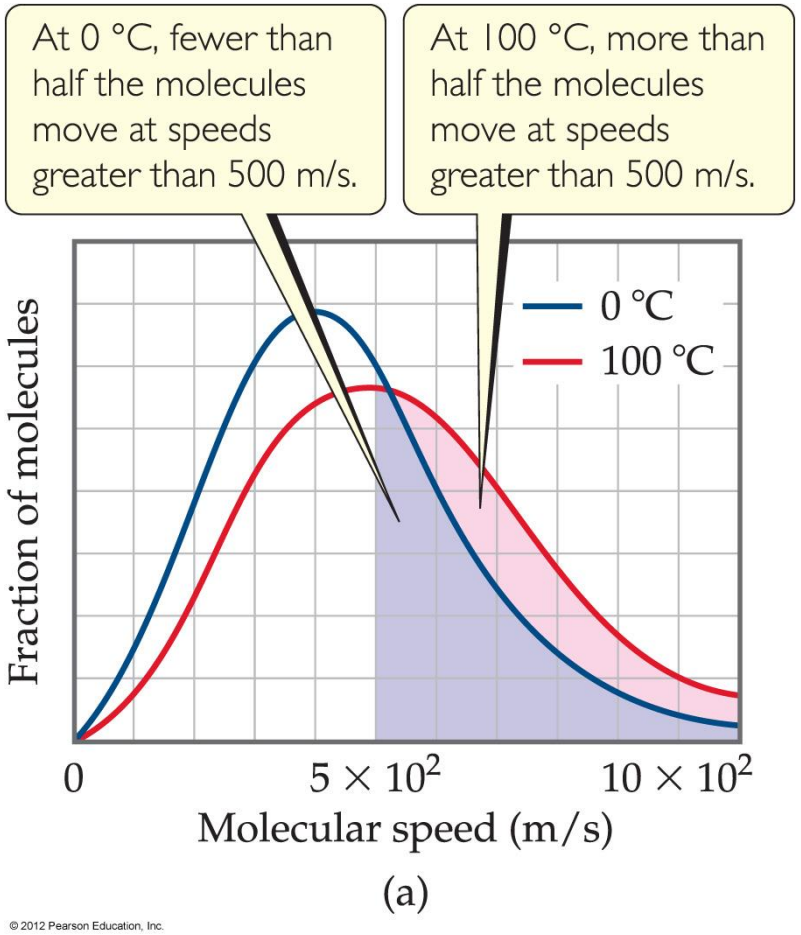
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Energy can be transferred between molecules during collisions, but the *average* kinetic energy of the molecules does not change with time, as long as the temperature of the gas remains constant.

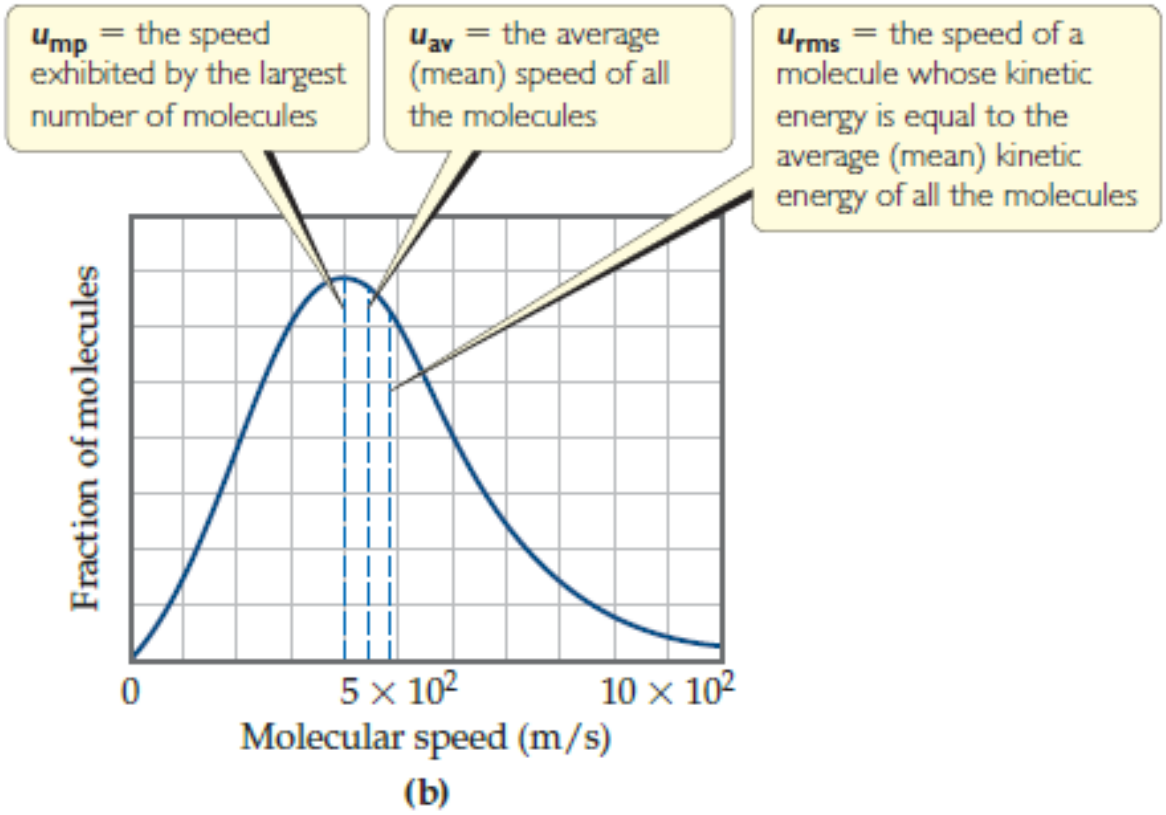
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Main Tenets of Kinetic-Molecular Theory

The average kinetic energy of the molecules is proportional to the absolute temperature.



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(b)

The rms speed is important because the average kinetic energy of the gas molecules in a sample is equal to $\frac{1}{2}m(u_{rms})^2$. (Section 5.1) Because mass does not change with temperature, the increase in the average kinetic energy $\frac{1}{2}m(u_{rms})^2$ as the temperature increases implies that the rms speed of the molecules (as well as u_{av} and u_{mp}) increases as temperature increases.

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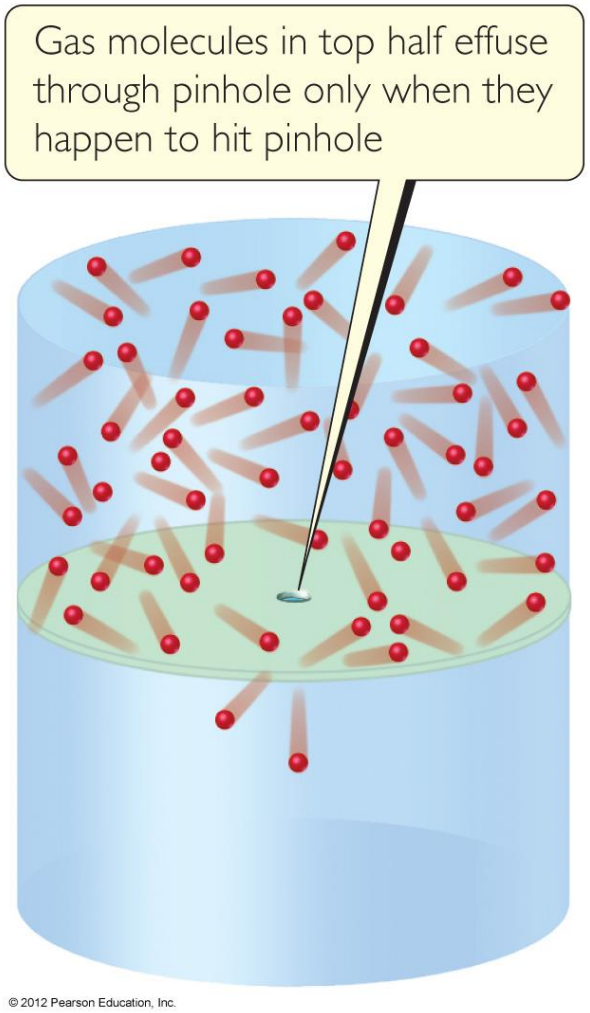
Application of Kinetic-Molecular Theory to the Gas Laws

The empirical observations of gas properties as expressed by the various gas laws are readily understood in terms of the kinetic-molecular theory. The following examples illustrate this point:

1. **An increase in volume at constant temperature causes pressure to decrease.** A constant temperature means that the average kinetic energy of the gas molecules remains unchanged. This means that the rms speed of the molecules remains unchanged. When the volume is increased, the molecules must move a longer distance between collisions. Consequently, there are fewer collisions per unit time with the container walls, which means the pressure decreases. Thus, kinetic-molecular theory explains Boyle's law.
2. **A temperature increase at constant volume causes pressure to increase.** An increase in temperature means an increase in the average kinetic energy of the molecules and in u_{rms}. Because there is no change in volume, the temperature increase causes more collisions with the walls per unit time because the molecules are all moving faster. Furthermore, the momentum in each collision increases (the molecules strike the walls more forcefully). A greater number of more forceful collisions means the pressure increases, and the theory explains this increase.

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Effusion

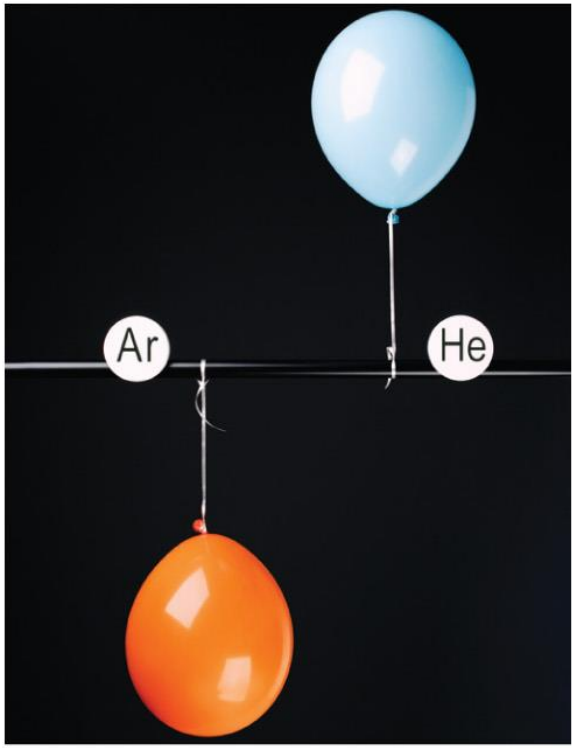


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Effusion is the escape of gas molecules through a tiny hole into an evacuated space.

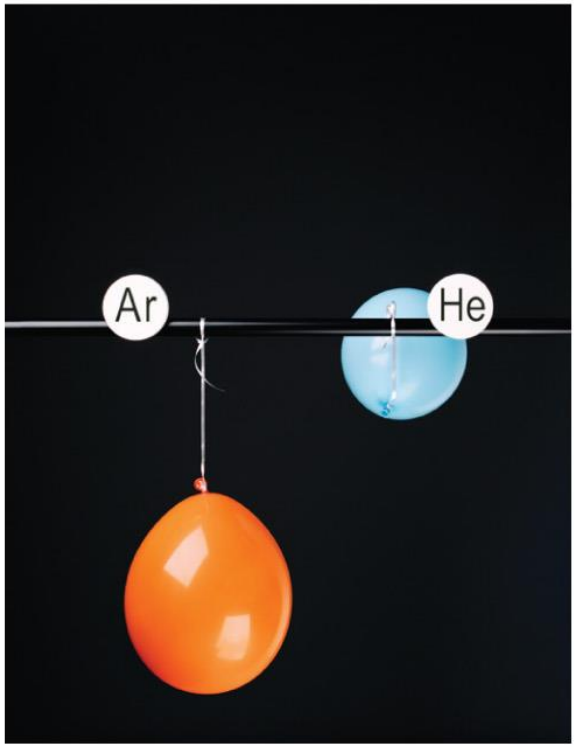
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Effusion



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Both gases effuse through pores in balloon, but lighter helium gas effuses faster than heavier argon gas

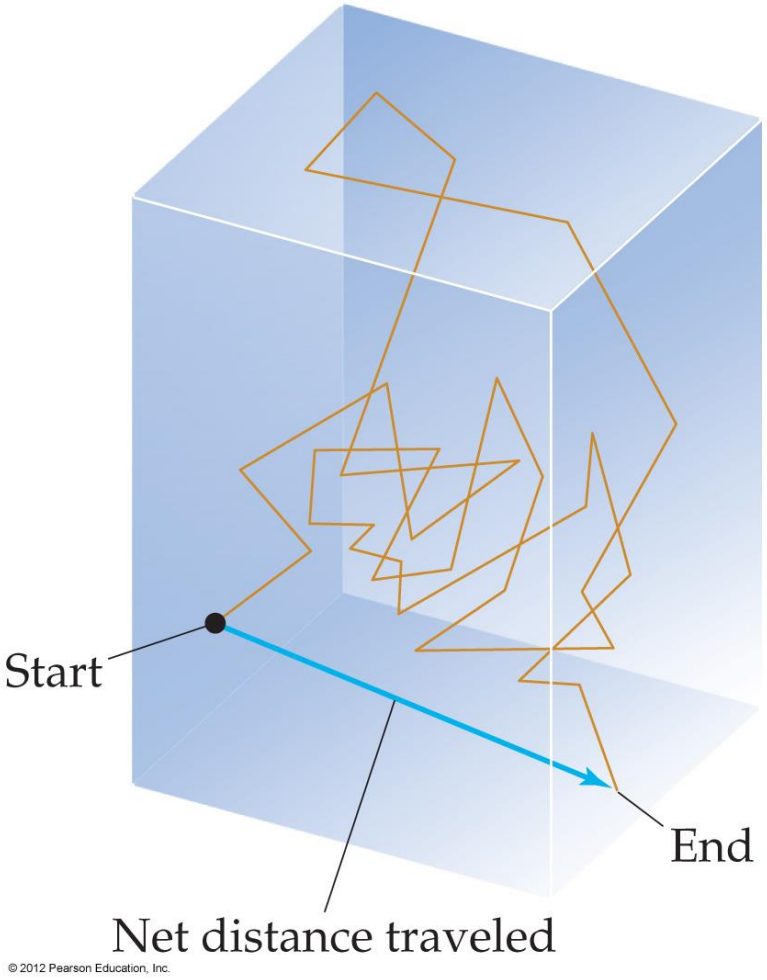


The difference in the rates of effusion for helium and nitrogen, for example, explains why a helium balloon would deflate faster.

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Diffusion

Diffusion is the spread of one substance throughout a space or throughout a second substance.



Gases

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10.8 | Molecular Effusion and Diffusion

According to the kinetic-molecular theory of gases, the average kinetic energy of *any* collection of gas molecules, $\frac{1}{2}m(u_{rms})^2$, has a specific value at a given temperature. Thus, for two gases at the same temperature a gas composed of low-mass particles, such as He, has the same average kinetic energy as one composed of more massive particles, such as Xe. The mass of the particles in the He sample is smaller than that in the Xe sample. Consequently, the He particles must have a higher rms speed than the Xe particles. The equation that expresses this fact quantitatively is

$$u_{rms} = \sqrt{\frac{3RT}{\mathcal{M}}} \qquad [10.20]$$

where $\mathcal{M}$ is the molar mass of the particles, which can be derived from the kinetic-molecular theory. Because $\mathcal{M}$ appears in the denominator, the less massive the gas particles, the higher their rms speed.

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SAMPLE EXERCISE 10.13 Calculating a Root-Mean-Square Speed

Calculate the rms speed of the molecules in a sample of N_2 gas at 25 °C.

SOLUTION

Analyze We are given the identity of a gas and the temperature, the two quantities we need to calculate the rms speed.

Plan We calculate the rms speed using Equation 10.20.

Solve We must convert each quantity in our equation to SI units. We will also use R in units of J/mol-K (Table 10.2) to make the units cancel correctly.

$$T = 25 + 273 = 298\ \text{K}$$

$$\mathcal{M} = 28.0\ \text{g/mol} = 28.0 \times 10^{-3}\ \text{kg/mol}$$

$$R = 8.314\ \text{J/mol-K} = 8.314\ \text{kg-m}^2/\text{s}^2\text{-mol-K} \quad (\text{Since } 1\ \text{J} = 1\ \text{kg-m}^2/\text{s}^2)$$

$$u_{\text{rms}} = \sqrt{\frac{3(8.314\ \text{kg-m}^2/\text{s}^2\text{-mol-K})(298\ \text{K})}{28.0 \times 10^{-3}\ \text{kg/mol}}} = 5.15 \times 10^2\ \text{m/s}$$

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Graham's Law of Effusion

In 1846, Thomas Graham (1805–1869) discovered that the effusion rate of a gas is inversely proportional to the square root of its molar mass. Assume we have two gases at the same temperature and pressure in two containers with identical pinholes. If the rates of effusion of the two gases are r_1 and r_2 and their molar masses are $\mathcal{M}_1$ and $\mathcal{M}_2$, **Graham's law** states that

$$\frac{r_1}{r_2} = \sqrt{\frac{\mathcal{M}_2}{\mathcal{M}_1}} \qquad [10.22]$$

a relationship that indicates that the lighter gas has the higher effusion rate.

The only way for a molecule to escape from its container is for it to "hit" the hole in the partitioning wall of Figure 10.15. The faster the molecules are moving the more often they hit the partition wall and the greater the likelihood that a molecule will hit the hole and effuse. This implies that the rate of effusion is directly proportional to the rms speed of the molecules. Because R and T are constant, we have, from Equation 10.22

$$\frac{r_1}{r_2} = \frac{u_{\text{rms1}}}{u_{\text{rms2}}} = \sqrt{\frac{3RT/\mathcal{M}_1}{3RT/\mathcal{M}_2}} = \sqrt{\frac{\mathcal{M}_2}{\mathcal{M}_1}} \qquad [10.23]$$

As expected from Graham's law, helium escapes from containers through tiny pinhole leaks more rapidly than other gases of higher molecular weight (Figure 10.16).

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Graham's Law

$$KE_1 = KE_2$$

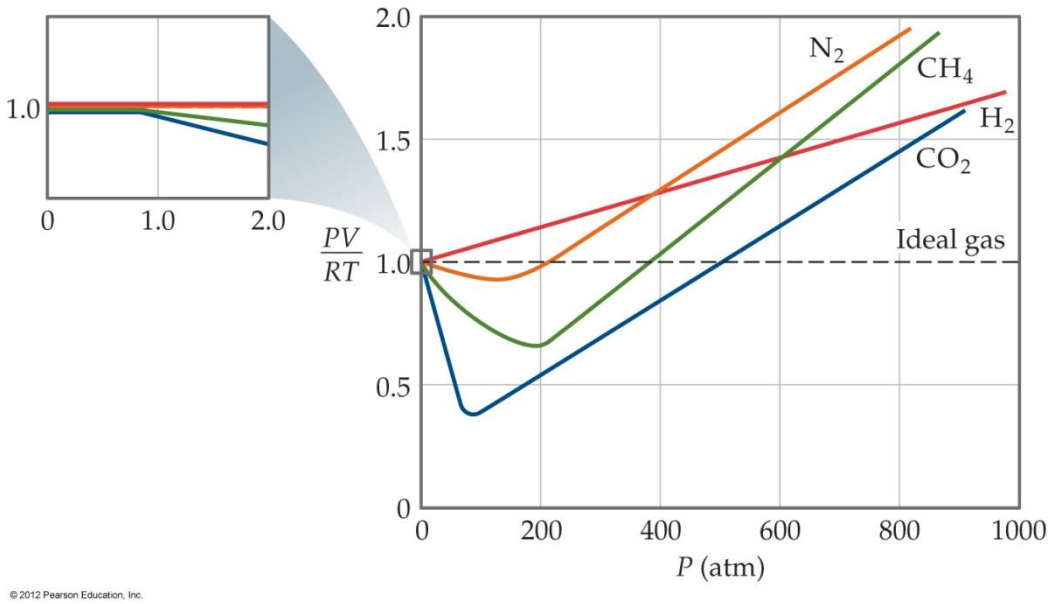
$$1/2\ m_1 v_1^2 = 1/2\ m_2 v_2^2$$

$$\frac{m_1}{m_2} = \frac{v_2^2}{v_1^2}$$

$$\frac{\sqrt{m_1}}{\sqrt{m_2}} = \frac{\sqrt{v_2^2}}{\sqrt{v_1^2}} = \frac{v_2}{v_1}$$

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Real Gases



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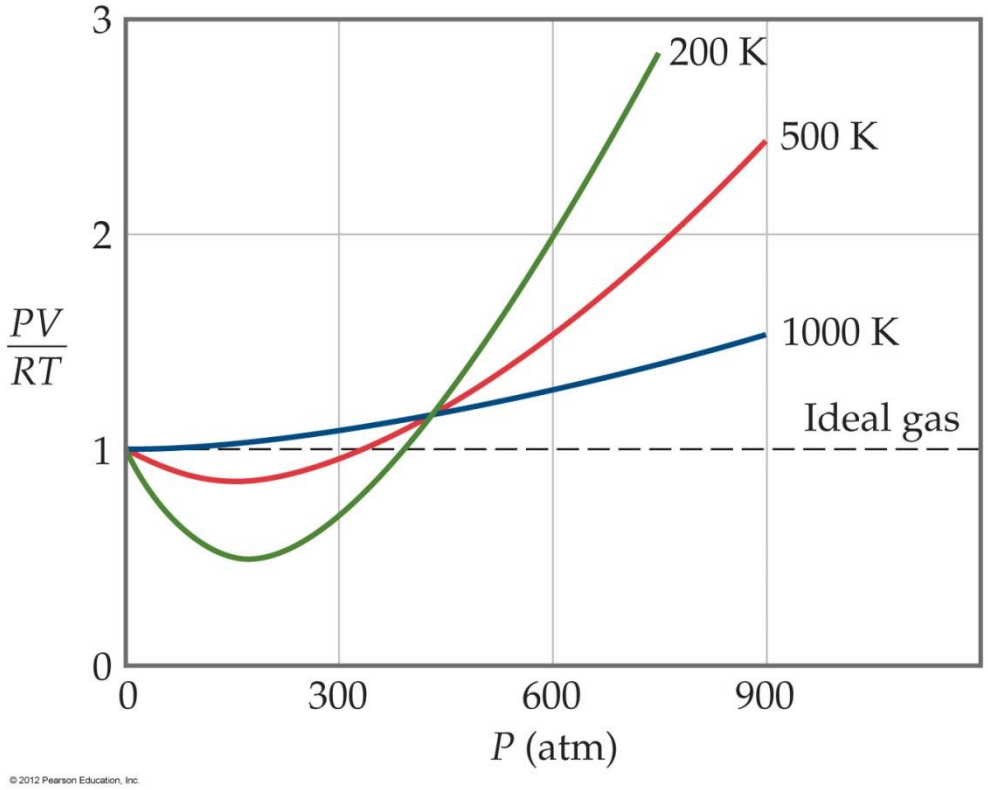
In the real world, the behavior of gases only conforms to the ideal-gas equation at relatively high temperature and low pressure.

Gases

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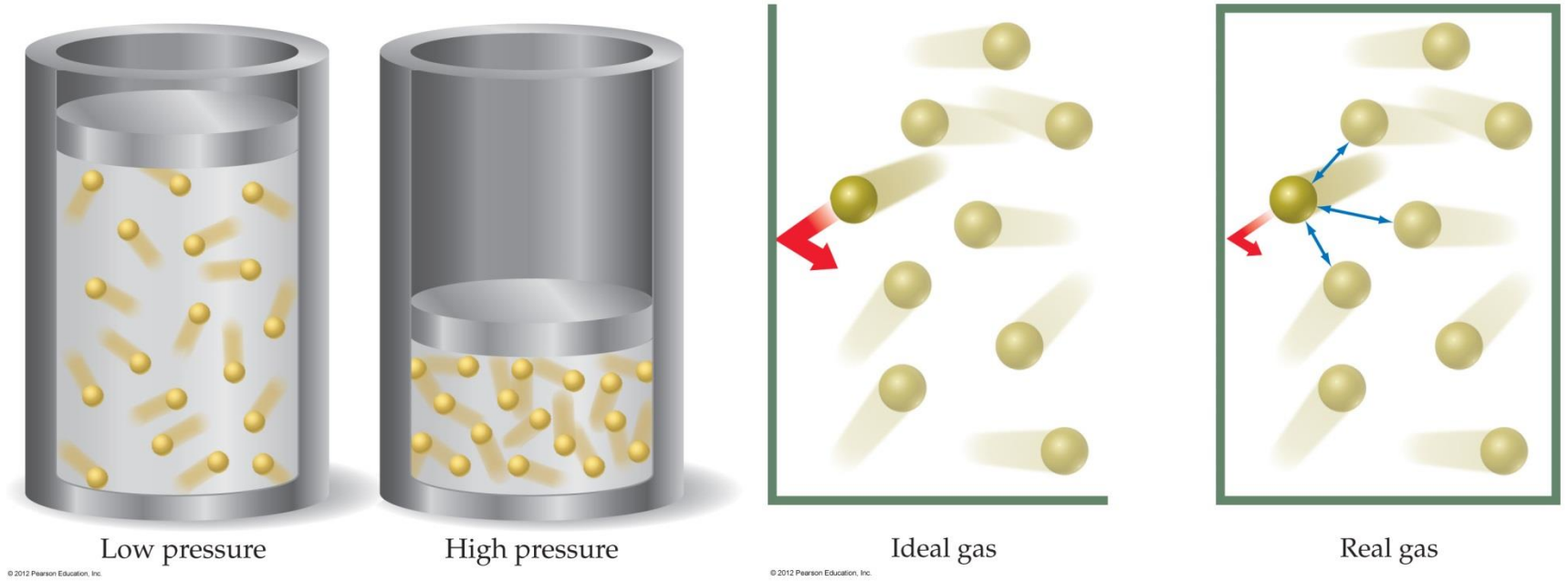
Real Gases

Even the same gas will show wildly different behavior under high pressure at different temperatures.



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Deviations from Ideal Behavior



The assumptions made in the kinetic-molecular model (negligible volume of gas molecules themselves, no attractive forces between gas molecules, etc.) break down at high pressure and/or low temperature.

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Corrections for Nonideal Behavior

- The ideal-gas equation can be adjusted to take these deviations from ideal behavior into account.
- The corrected ideal-gas equation is known as the **van der Waals equation**.

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The van der Waals Equation

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

TABLE 10.3 • Van der Waals Constants for Gas Molecules

Substance	a (L^2-atm/mol^2)	b (L/mol)
He	0.0341	0.02370
Ne	0.211	0.0171
Ar	1.34	0.0322
Kr	2.32	0.0398
Xe	4.19	0.0510
H_2	0.244	0.0266
N_2	1.39	0.0391
O_2	1.36	0.0318
Cl_2	6.49	0.0562
H_2O	5.46	0.0305
CH_4	2.25	0.0428
CO_2	3.59	0.0427
CCl_4	20.4	0.1383

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